STUTTER UNIT

User Operation Manual & Performance Guide • Daisy Seed Stereo DSP Engine

1. Welcome & Main Purpose

Welcome to the **Stutter Unit**! Whether you are performing live electronic sets, capturing glitchy rhythm breaks, or crafting ambient pitch-shifted textures, the Stutter Unit is designed to be your instant, expressive performance companion.

What does the Stutter Unit do?
At its core, the Stutter Unit is an *on-demand stereo audio buffer capture and micro-looping pedal*. While you play audio into the unit, it continuously monitors your incoming signal and tempo. The moment you press and hold the **Rate Encoder**, the pedal instantly grabs a clean stereo snippet of your performance (sliced cleanly to a musical subdivision like 1/16 or 1/8 note) and loops it indefinitely in real time.

While the loop is captured, turning the Rate Encoder sweeps the playback speed and pitch from sub-bass tape drops (0.25x) up to hyper-speed metallic glitches (4.0x). Releasing the encoder immediately drops you back into your live dry audio seamlessly! It gives you the intuitive control of hardware loopers like the Synthstrom Deluge with the power of high-fidelity stereo DSP.

Tip for Performers: *Use the Stutter Unit as a build-up riser during song transitions! Lock to MIDI clock, hit the stutter on a snare roll, and sweep the rate knob upward to create instant high-energy pitch risers before dropping back into the main beat.*

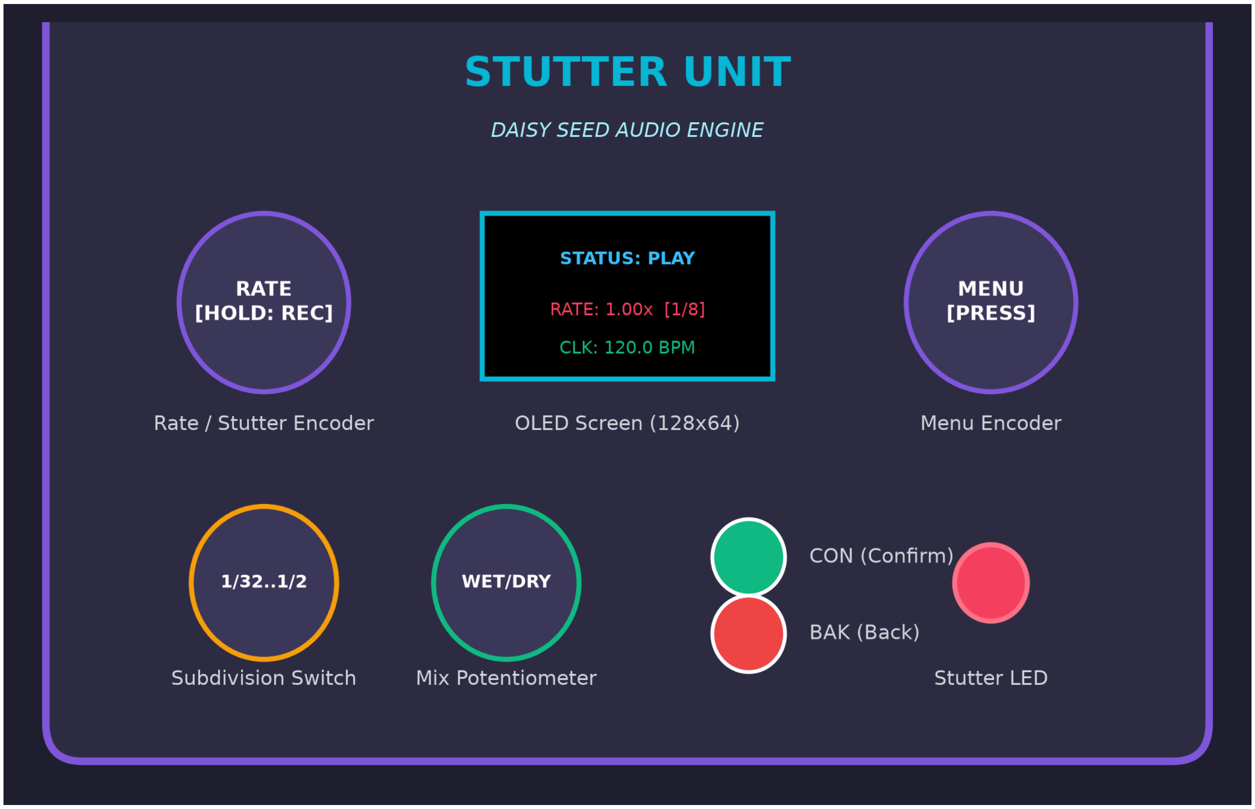
2. Signal Flow & Audio Architecture

Understanding how audio moves through the Daisy Seed audio engine:

Audio flows continuously through low-noise line input buffers into Daisy Seed SDRAM memory. When triggered, the DSP engine freezes the active subdivision window and plays it back through an ultra-smooth resampler before sending it to the analog Wet/Dry blend circuit.

3. Hardware Interface & Controls Overview

The physical layout puts all performance controls directly under your fingers:



Rate / Stutter Encoder	**Push & Hold:** Engages loop capture immediately. **Rotate:** Sweeps playback speed / pitch (0.25x to 4.0x).
5-Position Rotary Switch	Selects loop subdivision length: **1/32, 1/16, 1/8, 1/4, or 1/2 note**.
Wet / Dry Potentiometer	Adjusts the analog mix between your live dry signal and the stutter loop (0% to 100% Wet).
Menu Encoder	**Rotate:** Scroll OLED menu items. **Short Press:** Select / Edit parameter or confirm changes.
CON (Confirm) Button	Standalone tactile button acting as a redundant 'Enter' / Confirm key for menu navigation.
BAK (Back) Button	Standalone button to cancel menu edits, step back a screen, or exit the Debug view.
Stutter Indicator LED	Glows bright red whenever a stutter loop is active (recording or playing back).
OLED Display (128x64)	High-contrast SH1106 display showing play state, rate multiplier, subdivision, and MIDI clock.

4. Step-by-Step Operating Guide

4.1 Basic Momentary Stuttering

1. Set your desired loop length using the **5-Position Rotary Switch** (e.g., 1/8 note).
2. Turn the **Wet/Dry** knob to 12 o'clock (50/50 mix) or fully clockwise for 100% effect.
3. When you reach a spot in your audio you want to stutter, **press and hold the Rate Encoder**.
4. The **Stutter LED** lights up, and the OLED status changes from IDLE to PLAY.
5. Rotate the Rate Encoder left or right to bend the pitch down to 0.25x or up to 4.00x.
6. Release the Rate Encoder to stop looping and instantly resume dry audio playback.

4.2 OLED Screen & Menu Navigation

The system features four screen modes driven by an intuitive state machine:

Screen	Trigger / Event	Description
STATUS	Default Boot View	Displays live play state (IDLE/REC/PLAY), rate multiplier (e.g. 1.00x), subdivision, and MIDI clock BPM.
BROWSE	Press Menu Encoder or CON	Scroll through settings list (MIDI SYNC, QUANTIZE, PLAYBACK RATE MODE, DEBUG). Times out after 5s idle.
EDIT	Press CON on a Setting	Adjust parameter value using the Menu Encoder. Press CON to commit changes, or BAK to cancel.
DEBUG	Select DEBUG in Menu	Shows raw voltages, pin states, and switch values for hardware verification. Press BAK or CON to exit.

5. Advanced Quantization Modes

By default, turning the Rate Encoder produces continuous pitch sweeps (like changing tape speed). However, you can configure **Playback Rate Mode (PRM)** in the Settings Menu to quantize pitch to musical semitones!

Mode	Name	Behavior & Sound Character
OFF	Continuous (Default)	Smooth, unquantized rate sweeps from 0.25x to 4.00x. Ideal for classic tape stops and manual pitch bends.
LFQ	Loop-Frequency Quantized	Treats the loop length as a fundamental frequency oscillator. Rate snaps to musical semitones relative to loop length!
PTQ	Pitch-Detection Quantized	Runs an autocorrelation algorithm on captured audio to detect root pitch, then snaps knob sweeps to relative musical notes.

6. Complete MIDI Control Mapping

Connect a MIDI controller or DAW via TRS MIDI to remotely trigger or automate parameters:

Control	Target Variable	Msg Type	CC #	Value Translation & Behavior
Stutter Trigger	runtime.trigger_active	Note / CC	20	Note On = Rec/Play, Note Off = Idle | CC >= 64 = Active
Stutter Toggle	runtime.state	Note / CC	21	Toggles latching stutter state ON / OFF
Playback Rate	runtime.target_rate	CC / Pitch	22	0 = 0.25x, 64 = 1.0x (Unity), 127 = 4.0x
Semitone Offset	runtime.semitone_offset	CC / Pitch	23	0 = -24 semitones, 64 = 0 semitones, 127 = +24 semitones
Wet / Dry Mix	runtime.wet	CC	24	0 = 100% Dry, 127 = 100% Wet
Subdivision Switch	runtime.subdiv_pos	CC	25	0-25: 1/32, 26-50: 1/16, 51-76: 1/8, 77-102: 1/4, 103-127: 1/2
MIDI Sync Enable	config.midi_sync	CC	26	0-63 = OFF (Internal Tempo), 64-127 = ON (MIDI Clock)
Quantize Trigger	config.quantize	CC	27	0-63 = Immediate Trigger, 64-127 = Beat Quantized
Playback Rate Mode	config.prm_mode	CC	28	0-42 = OFF, 43-85 = LFQ, 86-127 = PTQ
Manual BPM	runtime.bpm	CC	29	0-127 mapped linearly to 40 - 240 BPM
Clear / Reset	Reset Routine	Note / CC	30	Instantly clears active buffer and returns to IDLE state
Bypass / Active	Audio Path Bypass	CC	31	0-63 = True Bypass, 64-127 = Effect Engaged

7. Troubleshooting & Hardware Reference

• No sound when holding Stutter?

Check Wet/Dry knob position and ensure audio input jacks are connected with 10µF DC-blocking capacitors.

• Buffer length out of beat sync?

Ensure **MIDI SYNC** is set to ON in the OLED menu and incoming MIDI clock (0xF8) is active.

• OLED Screen not displaying?

Verify I2C wiring (SCL to D11, SDA to D12) and 3.3V digital power supply connection.

• Enclosure Customization Graphics Placeholder

Below is a placeholder area for your physical enclosure artwork / drill template graphics.

[GRAPHICS / ENCLOSURE ARTWORK PLACEHOLDER]
Insert custom pedal faceplate photo, wiring diagram, or vector drill template graphics here.